P1-17 — Correction Log: Audit Corrections Applied to the Manuscript

Specification: P1_16_INDEPENDENT_AUDIT.md, items GAP-16-01 … GAP-16-08. **Files corrected:** P1_15_MANUSCRIPT_DRAFT.tex, P1_15_MANUSCRIPT_DRAFT.md (kept in sync; the LaTeX remains the version of record). **Not touched:** 02_THEOREM_BLUEPRINT.md (frozen; SHA-256 re-verified ec2fd8d58f8d2a986314f83997d750d689f7c88e98ec8f87753ab4c9cf438cff before and after editing), all proof packages (P1-04/05/06), P1_15_OPTIONAL_COROLLARIES.md (N1–N3 remain unadopted candidates, still not used in the manuscript), and every core theorem/corollary statement and proof idea. No novelty or priority wording was added anywhere.

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1. Bibliographic verification performed this phase (primary records actually inspected, 21 Oct 2026)

Entry | Primary record inspected | What it established

CDL2015 | Crossref record for DOI 10.1016/j.ffa.2014.11.006 (publisher-deposited) + ScienceDirect article page | Authors Bocong Chen Hai Q. Dinh Hongwei Liu; title ∗Repeated-root constacyclic codes of length 2ℓ^m p^n∗; FFA 33 (2015), 137–159.

Sharma2015 | Springer article page for DOI 10.1007/s12095-014-0106-5 | Anuradha Sharma, ∗Repeated-root constacyclic codes of length ℓ^t p^s and their dual codes∗, Cryptogr. Commun. 7 (2015), 229–255.

WZLL2023 | AIMS (publisher) article page for DOI 10.3934/amc.2021044 | Tingting Wu, Shixin Zhu, Li Liu, Lanqiang Li, ∗Repeated-root constacyclic codes of length 6ℓ^m p^n∗, AMC 17(5) (2023), 1154–1180.

ZWLL2025 | ScienceDirect article page for DOI 10.1016/j.ffa.2024.102542 + Crossref record | Qi Zhang, Weiqiong Wang, Shuyu Luo, Yue Li, ∗Repeated-root constacyclic codes of length k^sℓ^m p^n over finite fields∗, FFA 101 (2025), Art. 102542.

Talbi2021 | arXiv record 2102.06995 | Authors: Sarra Talbi, Aicha Batoul, Alexandre Fotue Tabue, Edgar Martínez-Moro; title as cited.

ES2026 (Eldin name) | ScienceDirect article page for DOI 10.1016/j.ffa.2025.102712 + arXiv record 2305.15237 (incl. submission history) | Publisher author form "Ramy Taki Eldin"; arXiv submitter string "Ramy Taki Eldin F." ⇒ full form **Ramy F. Taki Eldin**. The form "Ramy Abu Eldin" appearing in one internal audit table is unsupported by either primary record. The manuscript already used "R. F. Taki Eldin", so **no change was needed**; the conflict is hereby resolved in favour of the primary records.

FZ2017 (Galois inner product origin) | Springer article page for DOI 10.1007/s10623-016-0282-8; corroborated by multiple citing papers' abstracts | Yun Fan, Liang Zhang, ∗Galois self-dual constacyclic codes∗, Designs, Codes and Cryptography 84 (2017), 473–492 — the 2017 paper that introduced the ℓ-Galois inner product $\Sigma\, x_i\, y_i^{p^\ell}$, which is exactly the pairing of manuscript §2 with σ = p^k.

BDS2006 | Science journal page + arXiv record quant-ph/0610092 | Todd Brun, Igor Devetak, Min-Hsiu Hsieh, ∗Correcting quantum errors with entanglement∗, Science 314(5798) (2006), 436–439, DOI 10.1126/science.1131563 (DOI added to the entry).

ZKZ2026 (extra, see E12) | MDPI article page for DOI 10.3390/e28040407 | Enbin Zhang, Bo Kong, Xiying Zheng (initials supplied).

Nothing was filled in from memory: every field added above traces to a record in this table.

2. Edits applied (both manuscript versions unless stated)

- **E1 — Numbering (GAP-16-03; .tex only).** \newtheorem{remark}[theorem]{Remark} → \newtheorem∗{remark}{Remark}: remarks are now unnumbered, so the simple-root recovery is **Proposition 6.4** in the LaTeX, matching the Markdown companion exactly (which already had 6.4). A counter simulation over the corrected .tex (§4 below) confirms every other number is unchanged and identical to the companion (Thm 6.1/6.3, Prop 6.2, Cor 7.1–7.4, Thm 8.2/8.4, Examples 9.1–9.5 …).
- **E2 — Euclidean-dual lemma boundary cases (GAP-16-01; §4 Lemma).** Inserted, after the dimension sentence: the cases r = 0 (g = 1, C = K^n, C^⊥ = {0}, $h^\natural$ of degree n has no nonzero multiple of degree < n) and s = 0 (h = 1, C = {0}, C^⊥ = K^n, $h^\natural$ = 1) are verified from the definitions, then "Assume r, s ≥ 1 below, so that V is defined (and v = 0 gives d = 0)". This restores exactly the boundary handling present in the source proof P1-04 RR-3; no statement changed.

• **E3 — Constacyclic codes over A defined (GAP-16-02; §2).** Added: a λ-constacyclic code over A is an A-submodule of A^n closed under the componentwise λ-constacyclic shift $(c_0,\ldots,c_{n-1}) \mapsto (\lambda c_{n-1}, c_0, \ldots, c_{n-2})$; under $A^n \cong A[x]/(x^n-\lambda)$ the shift is multiplication by x, so these codes are exactly the ideals of $A[x]/(x^n-\lambda)$ — an equivalence needing no separability or compatibility — and "code over A" henceforth means such a code (cf. assembly Proposition 6.2 / P1-05 Prop. 3.1(ii)). Theorem 3.1's closing sentence now says "λ-constacyclic codes over A correspond to tuples of component assignments (Proposition …)" instead of "global codes (as A-submodules)".

• **E4 — Galois pairing attributed (GAP-16-05; §1 and §2).** New reference [FZ2017] added and cited at the first Introduction mention and at the §2 definition ("the k-Galois inner product introduced by Fan and Zhang"), with the code-first slot order unchanged.

• **E5 — EAQECC remark justified and kept conditional (GAP-16-04; §10).** The remark now states its hypotheses (N=1, $m_1=1$, [n,K] code over F_q, H = dim Hull_k(C)), defines c = rank($\Pi\rho(\Pi)^T$) for a parity-check matrix Π, and proves c = (n−K) − H: the kernel of the pairing map $D(C) \to (C^\perp)*$, $y \mapsto (z \mapsto \Sigma z_i y_i)$, is exactly C ∩ D(C). The parameter form [[n, K−H, d_EA; n−K−H]] is then attributed to the entanglement-assisted construction **in its Galois form** [DIMP2026] within the framework of [BDS2006], with d renamed d_EA = "the distance furnished by that construction and must be determined independently". The fencing is unchanged: the enumerator yields (K,H) only, no quantum parameter from it alone, affine case needs distance control.

• **E6–E10 — Bibliography completions (GAP-16-06a).** Titles and initials supplied from §1's records: CDL2015 (title), Sharma2015 (title), WZLL2023 (T. Wu, S. Zhu, L. Liu, L. Li + title), ZWLL2025 (Q. Zhang, W. Wang, S. Luo, Y. Li + title), Talbi2021 (S. Talbi, A. Batoul, A. Fotue Tabue, E. Martínez-Moro). BDS2006 gained its verified DOI.

• **E11 — Citation-scope narrowing (GAP-16-06c; §1).** "Galois hulls over chain and serial rings appear in [Talbi2021]" → "Galois hulls of cyclic serial codes over a finite chain ring are studied in [Talbi2021]" (the cited paper is the chain-ring one; no serial-ring reference was added, so the claim was narrowed instead).

• **E12 — Extra consistency (noticed during this phase).** ZKZ2026 authors given as surnames only; initials E. Zhang, B. Kong, X. Zheng supplied from the MDPI record (§1). Flagged here so the change is not silent.

• **E13 — Micro-clarification (GAP-16-07; Theorem 5.2).** Added "here and below, α_i refers to the orbit (s,O) under discussion" after the dimension displays.

• Also: a two-line header comment in the .tex records this correction round and points to this log. The author placeholder "[Author names and affiliations to be inserted]" is deliberately untouched (author-side item).

3. Verification performed

1. **Structural (on the corrected .tex):** braces balanced (depth scan), all environment types begin/end-matched (15, including math internals); 33 labels, all \ref targets defined; 16 \bibitems, **all cited, no undefined citations**; a counter simulation reproduces the intended numbering with Proposition 6.4 for the P=1 recovery and confirms no other number moved. The .md has 16 reference entries, [FZ2017] cited in its text, and no "serial rings" phrasing remains.

2. **Validators:** python3 code/validate_p1_14.py — exit 0, final line ALL P1-14 INDEPENDENT AUDIT CHECKS PASSED, output **byte-identical** to the saved code/validation_p1_14_output.txt (all four layers including the symbolic Layer A; sympy was already present in the environment — nothing was installed this phase). The manuscript's mathematics was not altered, so this confirms the corrected files still describe the validated statements; E2 restores the boundary handling already present in P1-04 RR-3, and E5's rank identity is proved inline in the remark itself (rechecked as item 4 below).

3. **Boundary-case re-verification (E2)** was done from the definitions, not by analogy: for r=0, $\langle h^\natural \rangle$ (multiples of degree < n, equivalently the ideal in the quotient, where $h^\natural = x^n - \lambda^{-1} \equiv 0$) is

{0} = C^⊥; for s=0, h^♮ = 1 generates K^n = C^⊥; in both cases the dimension count dim = r also remains valid, so the rest of the proof is untouched.

4. **Rank identity (E5)** checked by the kernel argument above; the matrix of the pairing map in the bases (rows of Π, rows of ρ(Π)) is precisely Π ρ(Π)^T, and dim D(C) = n − K.

5. **Frozen blueprint:** SHA-256 ec2fd8d5…cf438cff identical before/after; final git diff inspected to confirm only the intended files changed (manuscript .tex/.md, this log, gap register, worklog).

4. What could not be done / unresolved items

1. **LaTeX compilation:** no pdflatex/xelatex/latexmk/tectonic exists in this environment (checked this phase) and nothing was installed. The .tex is structurally verified as above but has **not** been compiled; compilation and a visual proofread remain author-side (this also covers display-equation numbering in the compiled output).

2. **Gate G1 / L1 (DPI2023 full text):** untouched and still DEFERRED; nothing in these corrections depends on it, and no sentence about L1 was strengthened.

3. **GAP-16-08:** the §1 characterizations of DIMP2026/ES2026 rest on the arXiv full texts (repo-verified); publisher versions were not theorem-checked — intentionally left as is.

4. **Author placeholder** remains for the authors to fill; journal-specific bibliography style (e.g. abbreviated journal names, issue numbers for FZ2017 84(3)) is a production-stage choice.

5. No mathematical issue remains open from the P1-16 audit; GAP-16-01…07 are addressed by E1–E13 and GAP-16-08 is a recorded no-action note.